

Yi-Chun Liao

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EDUCATION

Duke University

Ph.D. in Electrical and Computer Engineering

- Advisor: [Prof. Yiran Chen](#)

Durham, NC

Aug. 2026 – Present

National Taiwan University (NTU)

B.Sc. in Computer Science and Information Engineering (CSIE)

- Overall GPA: 3.90/4.30, Last 60 GPA: 4.09/4.30

Taipei, Taiwan

Sep. 2021 – Jan. 2026

PUBLICATIONS

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “RETENTION: Resource-Efficient Tree-Based Ensemble Model Acceleration with Content-Addressable Memory.” *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD 2026)*. [Link](#).

Yi-Chun Liao, Chieh-Lin Tsai, Yuan-Hao Chang, Camélia Slimani, Jalil Boukhobza, and Tei-Wei Kuo. “Practicalizing Tree-Based Model Acceleration with CAM through Model Pruning and Data Placement Optimization.” *ACM/IEEE International Conference on Hardware/Software Codesign and System Synthesis (CODES+ISSS 2025)*. [Link](#).

Hong-Wei Li, Ping-Ting Liu, Bo-Wei Lin, **Yi-Chun Liao**, and Yennun Huang. “IPMES: A Tool for Incremental TTP Detection Over the System Audit Event Stream.” *IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2024)*. [Link](#).

RESEARCH EXPERIENCE

Research Intern

University of Notre Dame, Advisor: [Prof. X. Sharon Hu](#)

Indiana, United States

May 2025 – Present

- Conduct independent research on realizing scalable in-memory retrieval with content-addressable memory (CAM), co-designing algorithm and hardware to prevent the massive data movement of retrieval phase.
- Conduct independent research on alleviating the capacity constraint of in-memory hyperdimensional computing, adapting the model for hardware constraints and introducing an algorithm for scalable CAM implementations.
- Contribute to the extension of the open-source CAM simulator [CAMASim](#), enabling data mapping optimizations with SPC and ODR, algorithms proposed in RETENTION.

Research Assistant

Academia Sinica, Advisors: [Prof. Yuan-Hao Chang](#) and [Prof. Tei-Wei Kuo](#)

Taipei, Taiwan

Apr. 2024 – Jun. 2026

- Conducted independent research on minimizing resource consumption for tree-based ensemble model inference, introducing an end-to-end framework [RETENTION](#) to minimize the capacity requirement for CAM-based acceleration.
- Explored and quantified the relationship between capacity redundancy and processing overhead, enabling flexible tradeoffs between controllable accuracy loss, energy consumption, and capacity requirement.
- Reduced capacity requirement by $4.35\times$ to $207.12\times$ with less than 3% accuracy loss and no degradation on throughput and energy efficiency. The first-authored paper was published in TCAD.
- Collaborated on minimizing retrieved tokens for RAG acceleration, exploring the tradeoffs between accuracy, latency, and capacity requirement.

Research Assistant

Academia Sinica, Advisor: [Prof. Yennun Huang](#)

Taipei, Taiwan

Jul. 2023 – Jun. 2024

- Collaborated on developing a data analytics tool, [IPMES](#), to detect structural anomalies and trace event patterns within system log streams. The co-authored paper was published in DSN 2024.
- Constructed an automated framework to simulate diverse system workloads and log operating system behavior inside virtual machines. The framework was subsequently used for dataset synthesis.

TEACHING EXPERIENCE

Teaching Assistant <i>Computer Architecture, Lecturer: Prof. Chia-Lin Yang</i>	Taipei, Taiwan <i>Sep. 2024 – Jan. 2025</i>
- Supported the instruction of the junior compulsory course Computer Architecture at NTU, enrolled by over 100 students. - Led weekly discussion sessions, designed and/or graded assignments, quizzes, and exams. - Provided voluntary office hours, exceeding 10 hours per week during exam periods.	
Teaching Assistant <i>APCS Camp</i>	Taipei, Taiwan <i>Jul. 2022/23 – Aug. 2022/23</i>
- Provided TA hours for over 300 C++ beginners during two-week crash courses. - Adapted teaching strategies including programming problem solving and customized course reviews to meet the diverse needs and skill levels of students.	

HONORS

First Place of Bachelor Thesis Award <i>National Taiwan University</i>	Taipei, Taiwan <i>2025</i>
Distinguished through a three-stage selection process and recognized as one of the top 2 in the CSIE Department, the best in the EECS College, and one of the top 6 university-wide.	
Appier Enterprise Award <i>National Taiwan University</i>	Taipei, Taiwan <i>2025</i>
Received the Most Potential Award from Appier in the departmental undergraduate research exhibition.	
Dean’s List Award <i>National Taiwan University</i>	Taipei, Taiwan <i>2025</i>
Ranked 1/155 in the department for both Spring and Fall 2024 semesters.	
Irving T. Ho Memorial Scholarship <i>Irving T. Ho Memorial Foundation</i>	Taipei, Taiwan <i>2024</i>
Awarded to 3 students in the department for academic and research excellence.	
First Place of AUO Enterprise Award <i>MakeNTU 2024 (Nationalwide Makerthon)</i>	Taipei, Taiwan <i>2024</i>
Designed an interactive in-car RPG game for AUO smart windows to enhance travel experience, which was recognized as the best work for AUO Enterprise Award among 43 teams.	

EXTRACURRICULAR ACTIVITIES

President <i>NTU CSIE Department Student Association</i>	Taipei, Taiwan <i>Aug. 2023 – Jul. 2024</i>
- Directed the student council of the CSIE department with over 50 members. - Promoted student welfare and represented over 600 students in faculty meetings, contributing to departmental decisions.	
Vice President <i>NTU EECS College Student Association</i>	Taipei, Taiwan <i>Aug. 2023 – Jul. 2024</i>
Represented over 1500 students in college faculty meetings and helped shape student-related policies.	
President <i>NTU Cancer Children Service Club</i>	Taipei, Taiwan <i>Feb. 2023 – Jul. 2023</i>
- Led a 50-member service club, managing budget allocation and arranging educational and recreational activities. - Worked with Childhood Cancer Foundation of R.O.C., organizing voluntary services to reduce the burden of parents.	
Vice Coordinator <i>NTU CSIE Camp</i>	Taipei, Taiwan <i>Feb. 2023 – Jul. 2023</i>
- Organized a 6-day camp for over 100 high school students with a 50-member staff team. - Designed activities to introduce students to the CSIE department and provide insights into university life.	